

U.S. Mineral Rankings

Dataset

Florian Geillon's *Mineral photos* dataset [1] contains over 39,000 high-quality photographs of the following minerals:

1. Azurite
2. Baryte
3. Beryl
4. Calcite
5. Cerussite
6. Copper
7. Fluorite
8. Gypsum
9. Hematite
10. Malachite
11. Pyrite
12. Pyromorphite
13. Quartz
14. Smithsonite
15. Wulfenite

Mineral grouping

As a preliminary exploration, we are evaluating three different models on three sets of minerals, grouped by their relevance to the current US administration. The United States Geological Survey (USGS) references the Energy Act of 2020 as defining critical minerals as “those that are essential to the economic or national security of the United States; have a supply chain that is vulnerable to disruption; and serve an essential function in the manufacturing of a product, the absence of which would have significant consequences for the economic or national security of the U.S.” [2]. The 2025 List of Critical Minerals includes from our list: Baryte, Beryl, Copper, and Fluorite; placing these in the high-priority group for our preliminary analysis.

Silicon is extracted from Quartz, so we place it in the high-priority set of minerals. Azurite and Malachite are not listed on the 2025 List of Critical Minerals, but are weathered copper ore, and are used as indicators of copper ore deposits [3]. Because of this, we placed them in the medium priority set of minerals. The medium priority set of minerals also includes: Smithsonite, a source of Zinc; Cerussite and Pyromorphite, sources of Lead. Wulfenite is then placed in the low priority set, as it is more defined as a Molybdenum ore, rather than a Lead source [4]. Also on the low priority set are Calcite, Gypsum, Hematite, and Pyrite, as they are not sources of Critical Minerals. The ranking of Critical Minerals is based on the USGS Risk Classifications [5].

Final groupings

High priority

- Baryte
- Beryl
- Copper
- Fluorite
- Quartz

Medium priority

- Azurite
- Malachite
- Smithsonite
- Cerussite
- Pyromorphite

Low priority

- Wulfenite
- Calcite
- Gypsum
- Hematite
- Pyrite

References

- [1] "Mineral photos." Accessed: Sep. 23, 2026. [Online]. Available: <https://www.kaggle.com/datasets/floriangeillon/mineral-photos>
- [2] "What is a critical mineral? | U.S. Geological Survey." Accessed: Sep. 23, 2026. [Online]. Available: <https://www.usgs.gov/faqs/what-a-critical-mineral>
- [3] "Azurite: The blue gem material, ore of copper, and pigment." Accessed: Sep. 24, 2026. [Online]. Available: https://geology.com/minerals/azurite.shtml#google_vignette
- [4] "Wulfenite Gemstone: Properties, Meanings, Value & More." Accessed: Sep. 24, 2026. [Online]. Available: https://www.gemrockauctions.com/learn/a-z-of-gemstones/wulfenite#google_vignette
- [5] "2025 Draft List of Critical Minerals | U.S. Geological Survey." Accessed: Sep. 24, 2026. [Online]. Available: <https://www.usgs.gov/media/images/2025-draft-list-critical-minerals>